

30903180103559-ShopSphere

ShopSphere Enterprise Production and Cloud Modernization

Digital Egypt Cubs Initiative - Level 5 Final Project | Student ID 30903180103559

Project links

Application	https://shopsphere-storefront.vercel.app
Review service	https://shopsphere-reviews.vercel.app/api
Repository	https://github.com/SSNABH/30903180103559-ShopSphere

Every link in this document opens without signing in to anything.

Supporting links

Main API	https://shopsphere-store-api.vercel.app/api
Health check endpoint	https://shopsphere-store-api.vercel.app/api/health
Uptime status page	https://stats.uptimerobot.com/LBmw2iOvH5
Scheduled cleanup job	https://shopsphere-jobs.vercel.app
Job, live execution	https://shopsphere-jobs.vercel.app/api/status
CI/CD pipeline runs	github.com/SSNABH/30903180103559-ShopSphere/actions

Test accounts

Administrator	admin@shopsphere.local	Admin123!
Customer	customer@shopsphere.local	Customer123!

Sign in as the administrator to reach the Admin Dashboard, where the store statistics include a review count fetched from the review service over REST.

Task 1 - Production Deployment

1.1 Frontend and backend both live on the public URLs above, with no build or runtime errors. Every one is checked by the pipeline's smoke job on each release.

1.2 PostgreSQL on Supabase, read and written by the deployed application. Registering an account, or adding an item to a cart, writes a row visible in the admin dashboard.

1.3 HTTPS, CORS, Helmet and rate limiting all active on the deployed backend. Rate limiting returns 429 after 20 requests to `/api/auth/login`.

1.4 Health endpoint above, monitored by UptimeRobot every 5 minutes, with the public status page linked above.

Task 2 - Cloud Preparation

2.1 [Architecture diagram](#)

2.2 [Cloud service classification](#)

2.3 [Namespace simulation evidence](#) | manifests in [k8s/](#)

Task 3 - Application Modernization

3.1 Review service deployed at its own URL above, from its own codebase in [review-service/](#) with its own dependencies and its own Vercel project. The review routes are gone from the main API: the reviews path returns 404 there and 200 on the review service.

3.2 Reviews on any product page are fetched from the review service.

3.3 [Serverless job evidence](#) - deployed, scheduled daily, and executing live at the status URL above.

3.4 [Architecture decision record](#)

Task 4 - Production Operations

4.1 [Pipeline runs](#) | [three environments](#) | main protected, requiring the pipeline to pass before a merge.

4.2 [Logging and operations](#)

4.3 [Rollback plan](#)

4.4 This document, also held in the repository as [30903180103559-ShopSphere.md](#).

A note on names

Every submitted document, the repository and this file carry the required **30903180103559-ShopSphere** name. The review service and the scheduled job are independently deployed codebases inside that repository, each with its own dependencies and its own Vercel project, so they are separate deployments living under the one required project name. The storefront carries the product name *DECI.Project*, the name it was built under in the first semester; the naming convention applies to the deliverables, not to the application's own branding.